

Binary codes

Binary codes are codes which are represented in binary system with modification from the original ones.

- ☐ Weighted Binary codes
- ☐ Non Weighted Codes

Weighted binary codes are those which obey the positional weighting principles, each position of the number represents a specific weight. The binary counting sequence is an example.

Decimal	BCD 8421	Excess-3	84-2-1	2421	5211	Bi-Quinary 5043210			5	0	4	3	2	1	0
0	0000	0011	0000	0000	0000	0100001		0		X					X
1	0001	0100	0111	0001	0001	0100010		1		X				X	
2	0010	0101	0110	0010	0011	0100100		2		X			X		
3	0011	0110	0101	0011	0101	0101000		3		X		X			
4	0100	0111	0100	0100	0111	0110000		4		X	X				
5	0101	1000	1011	1011	1000	1000001		5	X						X
6	0110	1001	1010	1100	1010	1000010		6	X					X	
7	0111	1010	1001	1101	1100	1000100		7	X				X		
8	1000	1011	1000	1110	1110	1001000		8	X			X			
9	1001	1111	1111	1111	1111	1010000		9	X		X				

Gray Code

The gray code belongs to a class of codes called minimum change codes, in which only one bit in the code changes when moving from one code to the next. The Gray code is non-weighted code, as the position of bit does not contain any weight. The gray code is a reflective digital code which has the special property that any two subsequent numbers codes differ by only one bit. This is also called a unit- distance code. In digital Gray code has got a special place.

Decimal Number	Binary Code	Gray Code	Decimal Number	Binary Code	Gray Code
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

Reflective Code

- A code is said to be reflective when code for 9 is complement for the code for 0, and so is for 8 and 1 codes, 7 and 2, 6 and 3, 5 and 4. Codes 2421, 5211, and excess-3 are reflective, whereas the 8421 code is not.

Sequential Codes

- A code is said to be sequential when two subsequent codes, seen as numbers in binary representation, differ by one. This greatly aids mathematical manipulation of data. The 8421 and Excess-3 codes are sequential, whereas the 2421 and 5211 codes are not.

Non weighted codes

Non weighted codes are codes that are not positionally weighted. That is, each position within the binary number is not assigned a fixed value. Ex: Excess-3 code

Excess-3 Code

Excess-3 is a non weighted code used to express decimal numbers. The code derives its name from the fact that each binary code is the corresponding 8421 code plus 0011(3).

Complements:

In digital computers to simplify the subtraction operation & for logical manipulation complements are used. There are two types of complements used in each radix system.

- i) The radix complement or r 's complement
- ii) The diminished radix complement or $(r-1)$'s complement

Representation of signed no.s binary arithmetic in computers:

- Two ways of rep signed no.s

1. Sign Magnitude form
2. 2. Complemented form

- Two complimented forms

1. 1's compliment form
2. 2. 2's compliment form

Advantage of performing subtraction by the compliment method is reduction in the hardware.(instead of addition & subtraction only adding circuit's are needed.) i. e, subtraction is also performed by adders only.

Instead of subtracting one no. from other the compliment of the subtrahend is added to minuend. In sign magnitude form, an additional bit called the sign bit is placed in front of the no. If the sign bit is 0, the no. is +ve, If it is a 1, the no is _ve.

Ex:

0	1	0	1	0	0	1
---	---	---	---	---	---	---



Sign bit

=+41 magnitude



1	1	0	1	0	0	1
---	---	---	---	---	---	---

= -41

Note: manipulation is necessary to add a +ve no to a -ve no

1's Complement Arithmetic

Steps:

- Convert the given numbers in 1's complement form.
- ADD them.
 - Carry: Add to the result (at the end)
 - No carry.
- MSB , If 0 ...check result is +vepure binary form
If 1, result is negative
Result is 1's complement

2's Complement Arithmetic

Steps:

- Convert the given numbers in 2's complement form.
- ADD them.
 - if Carry neglect them
- Check MSB , If 0 ...check result is +vepure binary form
If 1, result is negative
Result is 2's complement